

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

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Data engineer with hands-on experience building SQL-based data pipelines, data models, and automated data quality checks at Deutsche Bundesbank (AnaCredit), plus an independent end-to-end banking data pipeline project built with Claude Code as an AI-assisted development tool. Comfortable owning ETL/ELT logic and data modeling from ingestion through auditable documentation, with a data science background that adds statistical and analytical depth to engineering work.

Core Competencies

- **Data Engineering & ETL/ELT:** Pipeline design, data cleaning, referential integrity checks, SQL, SQLite, Pandas
- **Data Modeling & Quality:** Normalized schema design, migration testing, automated error detection, auditable data logs
- **Business Process Understanding:** Regulatory reporting workflows and business-process context for banking data (Deutsche Bundesbank AnaCredit)
- **Programming:** Python, SQL, Java, R, C++
- **AI-Assisted Development:** Daily use of AI coding tools (Claude Code) to accelerate pipeline development and automation
- **Cloud & Tools:** Azure Cognitive Services, AWS, Jira, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reported banking data
 - Automated Jira ticket creation and emails to reporting banks for detected data quality issues, significantly reducing response time

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts): normalized SQLite database with foreign-key integrity checks
 - Staged data cleaning (duplicates, nulls, date formats, fuzzy-matched categorical errors) with a fully auditable log recording table, column, old/new value, action, and timestamp
 - Automated account balance maintenance via SQL triggers; produced quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data, building data pipelines from raw records to predictive models and clustering
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data with high accuracy
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship).
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.